

Chapter

The Role of HER2 in Cancer: Molecular Mechanisms, Clinical Impact, and Advances in Targeted Therapies

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Abstract

Human Epidermal Growth Factor Receptor 2 (HER2) is a pivotal protein in cancer biology, primarily known for its role in breast cancer. As a receptor tyrosine kinase, HER2 is involved in regulating cell growth, differentiation, and survival. In many cancers, particularly breast cancer, HER2 is overexpressed or amplified, leading to more aggressive tumor growth and poorer patient outcomes. This chapter will explore the intricate molecular mechanisms by which HER2 contributes to oncogenesis, including its influence on signaling pathways that drive tumor progression. It will also examine the clinical significance of HER2 as both a biomarker and a therapeutic target. The impact of existing HER2-targeted treatments, such as trastuzumab and newer therapies, will be reviewed. Additionally, recent advancements in HER2 research will be discussed, along with future directions for overcoming resistance and improving treatment efficacy to enhance patient outcomes.

Keywords: molecular mechanisms, biomarker, therapeutic target, therapeutic approaches, prognostic implications

1. Introduction

The Epidermal Growth Factor Receptor (EGFR) family, also known as the ErbB receptor family, consists of four closely related transmembrane tyrosine kinase receptors: EGFR (ErbB1), ErbB2 (HER2), ErbB3, and ErbB4 [1]. These receptors play critical roles in regulating cellular processes such as growth, differentiation, survival, and repair (**Table 1**). They achieve this by mediating signal transduction pathways that are activated upon ligand binding, dimerization, and subsequent autophosphorylation of the receptor's intracellular tyrosine kinase domain [6]. Each member of the EGFR family has distinct and overlapping functions, dictated by their expression patterns and interactions with various ligands and co-receptors. Predominantly expressed in epithelial tissues, EGFR is a key regulator of cell proliferation and differentiation. Dysregulated EGFR signaling is associated with several cancers, including lung, colorectal, and head and neck cancers, where it

Receptor	Expression profile	Primary functions	Oncogenic dysregulation	Associated cancer types	Reference
EGFR (ErbB1)	Widely expressed in epithelial tissues	Regulates cell proliferation, differentiation, and repair	Overexpression, activating mutations (e.g., L858R, T790M)	Lung, colorectal, head and neck cancers	[2]
HER2 (ErbB2)	High in breast, gastric, and ovarian tissues	Acts as a co-receptor, amplifies signaling when dimerized	Gene amplification, overexpression	Breast, gastric, and ovarian cancers	[3]
ErbB3	Epithelial and neuronal cells	Signals via PI3K-AKT pathway, lacks intrinsic kinase activity	Heterodimerization with HER2	Breast cancer, therapy-resistant cancers	[4]
ErbB4	Neural and cardiac tissues	Involved in neurodevelopment and tissue differentiation	Dysregulated ligand binding or signaling	Breast cancer, neurodevelopmental disorders	[5]

Table 1.
EGFR family and associated oncogenic features.

promotes tumor growth, survival, and resistance to apoptosis [7]. Unlike other family members, HER2 does not directly bind ligands but acts as a preferred heterodimerization partner for other ErbB receptors, amplifying signaling potency [8]. Overexpression or gene amplification of HER2 is a hallmark of aggressive breast cancers and is associated with poor prognosis. This receptor lacks intrinsic kinase activity but forms heterodimers with other family members, particularly HER2 [9]. It is critical for signaling pathways like PI3K/AKT, which regulate cell survival and metabolism. ErbB3 dysregulation is implicated in resistance to targeted therapies in HER2-positive cancers. This receptor is unique in its ability to signal through both ligand-independent and ligand-dependent pathways. It is essential for neurodevelopment and tissue differentiation, with its dysregulation linked to certain cancers and neurological disorders [10]. The expression of these receptors is not restricted to epithelial tissues; they are also found in various other cell types, each contributing to distinct biological processes. ErbB2 and ErbB4 are involved in synaptic plasticity and neurodevelopment, influencing cognitive and sensory functions in neuronal cells [11]. EGFR and HER2 promote angiogenesis and vascular remodeling, processes that are co-opted in tumor progression in endothelial cells. EGFR expression in macrophages and lymphocytes modulates immune responses and inflammatory pathways in immune cells. ErbB receptors facilitate tissue repair and fibrosis, underscoring their role in wound healing and chronic inflammatory conditions in fibroblasts and mesenchymal cells [12]. HER2 stands out as a critical oncogenic driver in cancers, especially breast and gastric cancers [13]. Overexpression or gene amplification of HER2 leads to unchecked cellular proliferation, enhanced survival, and metastasis [14]. These characteristics underscore the necessity of targeting HER2 in cancer therapeutics. HER2-targeted therapies, including monoclonal antibodies (e.g., trastuzumab), tyrosine kinase inhibitors, and antibody-drug conjugates, have revolutionized the treatment landscape for HER2-positive cancers, significantly improving clinical outcomes [15]. This chapter delves into the molecular mechanisms underpinning HER2's role in cancer, its clinical significance, and the transformative advances in HER2-targeted therapies. By exploring these aspects, the chapter aims to provide a comprehensive understanding of HER2's multifaceted functions and its implications for cancer biology and therapy.

2. Functional biology of EGFR and its role in carcinogenesis

The EGFR family plays a central role in carcinogenesis, with its members regulating key cellular processes that, when dysregulated, contribute to tumor initiation, progression, and metastasis [16]. Ligand binding induces dimerization of EGFR family receptors, triggering intracellular signaling cascades such as the RAS-RAF-MEK-ERK pathway (promoting proliferation) and the PI3K-AKT-mTOR pathway (enhancing survival and growth) [17]. Dysregulation in these pathways, often due to mutations, overexpression, or gene amplification, underpins the oncogenic potential of EGFR family members. In normal cellular contexts, EGFR-mediated signaling is tightly regulated to ensure appropriate cellular responses. However, in cancer, aberrations such as EGFR mutations (e.g., EGFR exon 19 deletions or L858R mutations in non-small cell lung cancer) and HER2 amplification (commonly observed in breast and gastric cancers) lead to constitutive activation of these signaling networks [18]. This results in uncontrolled cell division, evasion of apoptosis, angiogenesis, and metastasis (**Figure 1**).

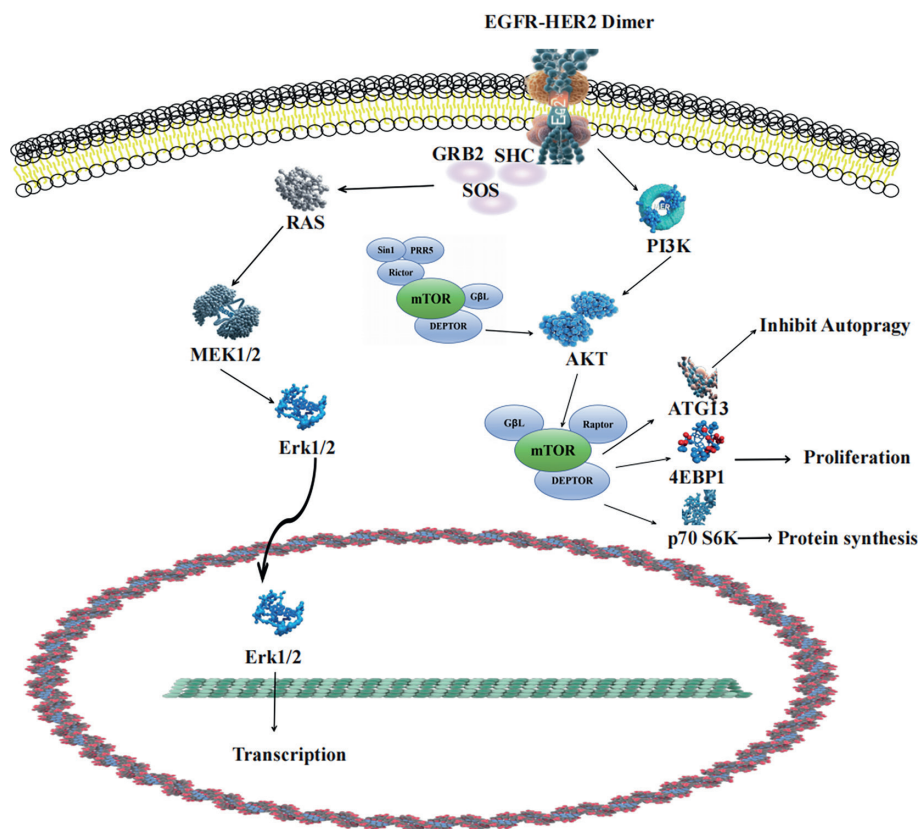


Figure 1. *EGFR-Mediated Signaling Pathways and Their Role in Carcinogenesis. The RAS-RAF-MEK-ERK pathway, emphasizing its role in promoting cell proliferation and migration. The PI3K-AKT-mTOR pathway, detailing its contribution to cell survival and growth.*

3. HER2's unique role in cancer biology and clinical significance

Human Epidermal Growth Factor Receptor 2 (HER2), a member of the EGFR family, distinguishes itself through its unique biological features and pivotal role in cancer progression [2]. Unlike other EGFR family members, HER2 lacks direct ligand-binding capability. Instead, it functions as a preferred heterodimerization partner, enhancing the signaling potency of other ErbB receptors [19]. This property amplifies critical pathways, such as PI3K-AKT and RAS-RAF-MEK-ERK, which regulate cell proliferation, survival, and metabolism [20]. The overexpression or gene amplification of HER2 is a hallmark of several aggressive cancers, including breast and gastric cancers [21]. This aberrant HER2 activity promotes uncontrolled cellular proliferation, enhanced survival, and increased metastasis, contributing to poor clinical outcomes. HER2's clinical importance is underscored by its dual role as a prognostic marker and a therapeutic target. In breast cancer, approximately 20–30% of cases exhibit HER2 overexpression or gene amplification, correlating with a more aggressive tumor phenotype characterized by rapid growth, higher metastatic potential, and worse prognosis if untreated [22]. Historically, HER2-positive breast cancers were associated with poor outcomes. However, the development of HER2-targeted therapies, including monoclonal antibodies (e.g., trastuzumab and pertuzumab),

tyrosine kinase inhibitors, and antibody-drug conjugates, has revolutionized treatment [23]. These therapies have significantly improved survival rates, transforming HER2-positive breast cancer from a highly aggressive disease to a manageable condition. Beyond breast cancer, HER2 plays a significant role in other malignancies. In gastric cancer, HER2 amplification or overexpression is observed in about 10–20% of cases [24]. Targeted therapies like trastuzumab have improved survival and treatment outcomes for these patients. HER2 involvement has also been identified in ovarian, endometrial, and non-small cell lung cancers (NSCLC), although the efficacy of HER2-targeted treatments in these cancers remains an active area of research [25]. The ability to identify HER2-positive tumors enables clinicians to tailor therapeutic strategies, improving patient outcomes and quality of life. HER2's broad relevance in oncology highlights its importance as both a biomarker and a therapeutic target across various cancer types.

4. The molecular architecture of HER2

HER2 is a tyrosine kinase receptor with a complex structure that consists of three major domains: the extracellular domain, the transmembrane region, and the intracellular tyrosine kinase domain (**Figure 2**). These components work together to mediate cell signaling.

4.1 Extracellular domain

The extracellular domain of HER2 is responsible for interacting with other ErbB family receptors (e.g., EGFR, HER3, and HER4) to form heterodimers [12]. Unlike other ErbB family receptors, HER2 does not bind a ligand directly but is always in a conformation ready to dimerize. Once HER2 forms a dimer with another receptor, this interaction triggers a conformational change that is transmitted through the transmembrane region to the intracellular tyrosine kinase domain [13].

4.2 Transmembrane region

The transmembrane region is a short hydrophobic sequence that anchors HER2 in the cell membrane [26]. It acts as a physical link between the extracellular domain and the intracellular tyrosine kinase domain. Upon dimerization at the

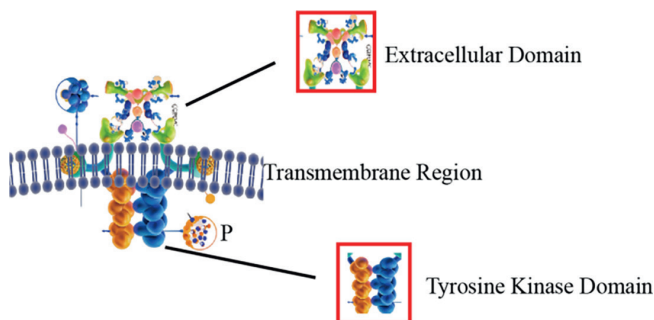


Figure 2.
HER2-HER3 dimer with a consists of three major domains.

extracellular domain, the transmembrane region helps transmit the activation signal across the cell membrane, initiating the activation of the intracellular tyrosine kinase domain [20].

4.3 Intracellular tyrosine kinase domain

The intracellular domain contains the tyrosine kinase activity, which is responsible for phosphorylating specific tyrosine residues on itself (autophosphorylation) and on other substrates [27]. This phosphorylation activates downstream signaling pathways, such as the PI3K/Akt and MAPK/ERK pathways, which regulate cell growth, survival, and differentiation [28]. Dimerization and the conformational changes from the extracellular and transmembrane regions activate the tyrosine kinase domain. Upon activation, the kinase domain catalyzes the transfer of phosphate groups to tyrosine residues on both HER2 and its dimerization partner, initiating intracellular signaling [29].

5. The varieties of dimers and their signaling specificities

5.1 Homodimers

Homodimers refer to a molecular complex formed when two identical molecules (monomers) bind together through non-covalent interactions, such as hydrogen bonds, hydrophobic interactions, or ionic bonds [30]. Homodimers are formed by two identical receptor molecules, such as EGFR-EGFR. These dimers primarily activate the MAPK pathway, promoting cell proliferation, differentiation, and survival in response to external stimuli [20].

5.2 Heterodimers

Receptor dimerization occurs when two EGFR family members come together to form a dimer [31]. This process is triggered by ligand binding to the extracellular domain of most family members, such as EGFR (ErbB1), HER3 (ErbB3), and HER4 (ErbB4). HER2, however, does not require a ligand due to its constitutively active dimerization interface [32]. Heterodimers consist of different EGFR family members, such as HER2-HER3 or HER2-EGFR. Heterodimers exhibit stronger and more sustained signaling compared to homodimers, owing to the complementary properties of the partnered receptors [9]. The HER2-HER3 heterodimer is one of the most potent signaling complexes in the EGFR family. It strongly activates the PI3K-AKT pathway, which is crucial for cell survival, metabolism, and resistance to apoptosis [33]. The HER2-EGFR dimer contributes to both the PI3K-AKT and MAPK pathways, supporting cell growth, survival, and potentially therapeutic resistance. The HER2-HER4 heterodimer influences pathways involved in cell differentiation and tissue development [10].

6. The functional ramifications of dimerization

In healthy cells, HER2 (human epidermal growth factor receptor 2) plays a vital role in regulating normal cell growth, differentiation, survival, and repair. Unlike other members of the ErbB family, HER2 is constitutively active and does not require

ligand binding for activation [34]. It serves as a preferred partner for other ErbB receptors, such as EGFR (HER1), HER3, and HER4. The ability of HER2 to form heterodimers with these receptors leads to the activation of downstream signaling pathways that regulate key cellular processes. Heterodimers involving HER2 generate stronger and more sustained signaling compared to homodimers, ensuring a robust cellular response [35]. Upon dimerization, the intracellular tyrosine kinase domains of HER2 are activated, leading to autophosphorylation of specific tyrosine residues. This autophosphorylation of HER2 triggers several critical signaling pathways that regulate essential cellular functions. The PI3K/Akt pathway plays a central role in promoting cell survival by inhibiting apoptosis (programmed cell death) and is also involved in regulating metabolism, growth, and protein synthesis, ensuring that cells can survive under stressful conditions [36]. The MAPK pathway is crucial for cell proliferation, differentiation, and survival, driving appropriate cell division and growth in response to external signals and maintaining tissue integrity. Additionally, the JAK/STAT pathway, activated by HER2, is important for immune responses, cell division, and differentiation, and helps regulate the balance between cell proliferation and apoptosis in various tissues, contributing to cellular homeostasis [15]. Together, these pathways ensure that HER2 signaling plays a vital role in controlling cell growth, survival, and tissue integrity under normal conditions. In normal cells, HER2 signaling is tightly regulated to ensure that cell growth, differentiation, and repair occur in a controlled manner [37]. This regulation helps maintain tissue homeostasis, allowing tissues to repair and regenerate while preventing excessive or uncontrolled growth. During embryonic development and adult tissue maintenance, HER2 plays a key role in organ formation and maintaining tissue integrity, contributing to the development of the nervous system, heart, and skin [38]. In healthy cells, the expression and activity of HER2 are carefully regulated to balance cellular responses to growth factors and other external signals. This ensures appropriate cell growth, survival, and differentiation, facilitating tissue repair and adaptation to changing conditions [39].

However, in cancerous conditions, the regulatory mechanisms controlling HER2 activity are often disrupted. Overexpression or amplification of HER2 occurs in many cancers, including breast, gastric, and ovarian cancers, due to genetic alterations that increase HER2 receptor levels on the cell surface [3]. Elevated HER2 levels enhance the likelihood of dimerization even in the absence of ligands, leading to continuous activation of downstream signaling pathways. This sustained signaling drives tumorigenesis and cancer progression by promoting uncontrolled cell growth and survival [40]. The overactivation of key pathways, such as the MAPK and PI3K/Akt pathways, results in excessive cell proliferation and resistance to apoptosis. Cancer cells become more resistant to programmed cell death by continuously activating survival pathways and suppressing pro-apoptotic signals, enabling them to thrive in hostile environments, such as nutrient deprivation or immune attack [41]. Additionally, HER2 overexpression promotes cancer cell invasion and metastasis. It triggers the epithelial-mesenchymal transition (EMT), which alters cell adhesion and motility, allowing tumor cells to invade surrounding tissues and spread to distant organs [42]. HER2 also interacts with stromal cells, immune cells, and components of the extracellular matrix to create a pro-metastatic niche, supporting tumor growth and dissemination. Furthermore, HER2 dysregulation upregulates the production of pro-angiogenic factors, such as vascular endothelial growth factor (VEGF), which stimulate the formation of new blood vessels to supply the tumor with nutrients and oxygen, facilitating rapid tumor growth [43]. This angiogenesis further enhances the tumor's ability to expand and metastasize, creating a vicious cycle that drives cancer progression.

7. The progress in HER2-targeted therapies

The discovery of HER2 as a prognostic marker has revolutionized the treatment of various cancers, particularly HER2-positive breast cancer. HER2-positive tumors, which account for 15–20% of breast cancer cases, were historically associated with poor outcomes due to their aggressive nature, rapid progression, and high metastatic potential [44]. HER2 overexpression drives tumor growth by continuously activating key growth signaling pathways, such as PI3K/Akt and MAPK, which promote cell division, survival, and resistance to apoptosis [45]. These pathways contribute to the rapid proliferation and greater metastatic potential of HER2-positive tumors. Beyond breast cancer, HER2 amplification is found in 10–30% of gastric and esophageal cancers, particularly in gastroesophageal junction adenocarcinomas, where it similarly contributes to aggressive disease and a poor prognosis [46]. Although less frequently observed, HER2 amplification is also present in other cancers, such as ovarian, bladder, and non-small cell lung cancer (NSCLC), where it correlates with more aggressive disease and unfavorable clinical outcomes. However, the advent of HER2-targeted therapies has transformed the landscape for HER2-positive cancers, improving patient outcomes and providing new treatment options [47]. Trastuzumab (Herceptin), the first major breakthrough in HER2-targeted therapy, specifically binds to the extracellular domain of the HER2 receptor, blocking its signaling and slowing tumor growth. Trastuzumab marks tumor cells for destruction by the immune system through antibody-dependent cellular cytotoxicity (ADCC) [48]. It has become the standard of care for HER2-positive metastatic gastric cancer and gastroesophageal junction adenocarcinoma, as well as HER2-positive breast cancer. Other HER2-targeted therapies have since emerged, including pertuzumab (Perjeta), ado-trastuzumab emtansine (Kadcyla), trastuzumab deruxtecan (Enhertu), margetuximab (Margenza), and fam-trastuzumab deruxtecan-nxki (Enhertu) [49]. These monoclonal antibodies function by binding to HER2, preventing receptor dimerization, and activating ADCC, ultimately inhibiting tumor growth. Each of these therapies represents a significant step forward in cancer treatment, enabling more targeted interventions that specifically block HER2 signaling while reducing off-target effects.

The development of these therapies has dramatically improved the prognosis for patients with HER2-positive cancers, shifting them from a once-dismal outlook to a more manageable condition with the promise of personalized treatment options. As a result, HER2 status has become a critical factor in guiding therapeutic decisions, optimizing treatment strategies, and improving patient outcomes [50]. These HER2-targeted therapies are often used in combination with other treatments, such as, to maximize their effectiveness. For example, trastuzumab is commonly used alongside chemotherapy in HER2-positive breast cancer, while newer treatments like trastuzumab deruxtecan (Enhertu) and pertuzumab are incorporated into combination regimens to overcome resistance and improve efficacy [51]. The success of HER2-targeted therapies has expanded the treatment options available for HER2-positive cancers, particularly in breast, gastric, and other HER2-overexpressing tumors [23]. The continuous development and refinement of these therapies, supported by ongoing clinical research and collaboration between biotechnology companies, have transformed the therapeutic landscape, offering hope to patients and improving survival rates. **Table 2** below organizes these therapies based on their class (monoclonal antibodies vs. small molecule inhibitors), outlining their mechanisms of action, indications, and manufacturers, providing a comprehensive overview of the current and emerging treatments for HER2-positive cancers.

Therapy	Class	Mechanism of Action	Indications	Manufacturer	Reference
Trastuzumab (Herceptin)	Monoclonal Antibody	Binds to the extracellular domain of HER2, inhibiting dimerization and signaling; promotes ADCC	HER2-positive breast cancer, gastric cancer, others	Genentech (Roche)	[21]
Pertuzumab (Perjeta)	Monoclonal Antibody	Binds to a different site on HER2, preventing receptor dimerization with HER3	HER2-positive breast cancer, neoadjuvant and metastatic	Genentech (Roche)	[23]
Ado-trastuzumab emtansine (Kadcyla)	Antibody-Drug Conjugate (ADC)	Trastuzumab conjugated to emtansine (DM1), delivering cytotoxic chemotherapy directly to HER2-positive cells	HER2-positive breast cancer, metastatic	Genentech (Roche)	[52]
Trastuzumab deruxtecan (Enhertu)	Antibody-Drug Conjugate (ADC)	Trastuzumab linked to deruxtecan (topoisomerase inhibitor), targeting HER2-positive cells and delivering cytotoxic agent	HER2-positive breast cancer, gastric cancer, others	Daiichi Sankyo, AstraZeneca	[53]
Margetuximab (Margenza)	Monoclonal Antibody	Engineered antibody with enhanced Fc receptor binding, improving immune response against HER2-positive cells	HER2-positive breast cancer (advanced, refractory)	MacroGenics	[54]
Fam-trastuzumab deruxtecan-nxki (Enhertu)	Antibody-Drug Conjugate (ADC)	Delivers topoisomerase inhibitor (deruxtecan) directly to HER2-positive cells	HER2-positive breast cancer, gastric cancer, others	Daiichi Sankyo, AstraZeneca	[55]
Lapatinib (Tykerb)	Small Molecule Inhibitor	Dual inhibitor of HER2 and EGFR tyrosine kinases, preventing receptor activation and downstream signaling	HER2-positive breast cancer (with chemotherapy)	GlaxoSmith Kline (GSK)	[56]
Neratinib (Nerlynx)	Small Molecule Inhibitor	Irreversible pan-HER inhibitor targeting HER1, HER2, and HER4, inhibiting signaling pathways	Extended adjuvant treatment of HER2-positive breast cancer	Puma Biotechnology	[57]
Afatinib (Gilotrif)	Small Molecule Inhibitor	Irreversible inhibitor of EGFR family receptors, including HER2, blocking tumor growth signals	HER2-mutated NSCLC	Boehringer Ingelheim	[58]
Tucatinib (Tukysa)	Small Molecule Inhibitor	Selective HER2 inhibitor, prevents HER2 signaling and tumor cell proliferation	HER2-positive breast cancer (including brain metastasis)	Seattle Genetics (Seagen)	[59]

Table 2.
HER2-Targeted Therapies.

8. The resistance to HER2-targeted therapies

HER2 amplification can lead to resistance to HER2-targeted therapies, presenting a significant challenge in cancer treatment. Resistance mechanisms are multifaceted and often involve alterations at the molecular level that enable tumor cells to evade the effects of these therapies [2]. One common mechanism is the development of mutations in the HER2 receptor itself, which can alter the binding site for drugs like trastuzumab, reducing their efficacy. Additionally, tumors may activate alternative growth signaling pathways, such as those mediated by the PI3K/Akt, MAPK, or MET pathways, which bypass HER2 dependency and sustain tumor proliferation and survival [60]. Changes in the tumor microenvironment can also contribute to resistance, including alterations in stromal interactions, immune evasion, or the production of growth factors that diminish the effectiveness of HER2-targeted agents. For example, increased expression of ligands that activate HER3 or EGFR can enhance compensatory signaling, while upregulation of efflux pumps may reduce intracellular drug concentrations, particularly for antibody-drug conjugates [61]. In gastric and esophageal cancers, resistance to HER2-targeted therapies is similarly observed, driven by mechanisms such as HER2 receptor truncation, which eliminates the binding site for monoclonal antibodies, or a switch in tumor phenotype, such as epithelial-to-mesenchymal transition (EMT), which reduces HER2 expression [25].

9. The approaches to overcome resistance to HER2-targeted therapies

To overcome this resistance, various approaches are being explored. Combination therapies that target multiple pathways simultaneously, for example, pairing HER2 inhibitors with drugs that block other relevant kinases, are being investigated [62]. Lapatinib, a HER2 and EGFR dual inhibitor, has been combined with mTOR inhibitors like everolimus in clinical trials to target both HER2 and the activated PI3K/AKT/mTOR pathway [63]. Another strategy is the development of novel HER2-targeted agents with enhanced binding affinity or the ability to overcome specific resistance mutations. ADCs (antibody-drug conjugates) like ado-trastuzumab emtansine (T-DM1) have shown promise [59]. T-DM1 combines the HER2-targeting antibody trastuzumab with a cytotoxic agent, allowing for more effective delivery of the drug to HER2-expressing cancer cells [64]. Moreover, research is focused on understanding the role of the tumor microenvironment in promoting resistance and devising ways to modulate it to enhance the efficacy of HER2-targeted therapies. For example, studies have shown that fibroblasts in the tumor stroma can secrete factors that activate HER2-independent survival pathways in cancer cells [65]. Strategies to target these stromal cells or block the secreted factors are being explored. Overall, continuous efforts in deciphering the molecular mechanisms and devising innovative approaches are essential to improve the outcomes of patients receiving HER2-targeted therapies.

10. Conclusion

While the advent of HER2-targeted therapies has significantly improved outcomes for patients with HER2-positive cancers, ongoing research seeks to address challenges such as resistance and disease recurrence. Emerging strategies include bispecific antibodies, which simultaneously target HER2 and other receptors, and the use of

immune checkpoint inhibitors in combination with HER2 therapies to harness the immune system's ability to fight cancer. Additionally, liquid biopsy techniques are being developed to monitor HER2 expression dynamically throughout treatment, providing a less invasive method for detecting resistance or disease progression. As more is learned about the complex biology of HER2, these advances will likely lead to further refinement of therapeutic approaches, improving patient outcomes even more. In conclusion, HER2 plays a central role in the pathogenesis of various cancers. Its discovery as a targetable oncogene has transformed cancer treatment, particularly in breast cancer, where HER2-targeted therapies have become the standard of care. While challenges such as resistance remain, the ongoing development of new therapies and treatment strategies continues to push the boundaries of cancer care, offering hope for improved survival and quality of life for patients with HER2-positive cancers.

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